

ODM-005 — Alternative Explanations

Status: Foundational

Authority: Methodology

Depends on:

- ODM-000 through ODM-004
 - OCS-000 through OCS-032
 - OAS-000 through OAS-031
 - OIS-000 through OIS-019
 - OGD-000 through OGD-023
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Purpose

This specification defines the constitutional role of Alternative Explanations within the Organodynamics Discovery Methodology.

Before proposing a preferred explanation, a researcher SHALL actively consider multiple plausible explanations for the same observed phenomenon.

The purpose of Alternative Explanations is not to delay discovery.

It is to strengthen discovery by protecting it from premature certainty.

Methodological rigor SHALL favor exploration before commitment.

1. Objective

The objective of Alternative Explanations is to ensure that every proposed discovery is evaluated within a landscape of plausible competing interpretations.

The process SHALL preserve:

Intellectual openness.

Methodological neutrality.

Comparative reasoning.

Historical accountability.

Evidence-based evaluation.

2. Definition

An Alternative Explanation is a constitutionally plausible account of a Problem Frame that differs meaningfully from other proposed accounts while remaining consistent with the available observations and recognized patterns.

Alternative explanations MAY be mutually incompatible.

Their coexistence SHALL be explicitly documented until sufficient evidence justifies preferring one over another.

3. Generation of Alternatives

Researchers SHOULD generate multiple alternative explanations whenever constitutionally feasible.

Each explanation SHALL identify:

Its underlying assumptions.

The observations it explains.

The observations it does not yet explain.

Its predicted consequences.

Its relationship to existing constitutional knowledge.

Generation SHALL prioritize diversity of reasoning rather than quantity.

4. Comparative Evaluation

Alternative explanations SHALL be compared using explicit criteria, including:

Explanatory power.

Consistency with observations.

Conceptual economy.

Compatibility with the existing constitutional corpus.

Predictive potential.

Recognition of unresolved questions.

Evaluation SHALL remain transparent and reproducible.

5. Relationship to Hypothesis

A formal Hypothesis SHALL identify the Alternative Explanations that were considered during its development.

Rejected alternatives SHALL remain part of the permanent methodological record.

Future evidence MAY justify their reconsideration.

6. Historical Preservation

Every Alternative Explanation SHALL preserve:

Its original formulation.

Supporting observations.

Reasons for acceptance or rejection.

Subsequent revisions.

Historical status.

Methodological history SHALL remain discoverable.

7. Verification

Verification SHALL demonstrate:

Transparent comparison.

Evidence traceability.

Conceptual fairness.

Historical continuity.

Independent reproducibility.

The process SHALL remain explainable.

8. Alternative Explanations Test

Alternative Explanations satisfy this specification when independent reviewers consistently identify and compare substantially equivalent sets of plausible explanations for the same Problem Frame using the available evidence, without prematurely converging on a single preferred interpretation.

Conformance

A methodological investigation conforms to this specification when it documents multiple constitutionally plausible explanations, evaluates them transparently, preserves their historical record, and maintains clear separation between comparative exploration and subsequent hypothesis selection.

Methodological Principle

The first explanation is often persuasive.

It is not therefore the best explanation.

Organodynamics protects discovery by requiring every promising idea to stand beside its strongest alternatives before it is allowed to stand alone.

Progress begins not when certainty appears,

but when competing explanations are examined with equal intellectual honesty.

End of Specification ODM-005